

C978

Science Methods—Secondary Earth Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

An earth science teacher with a long hallway and a roll of adding-machine tape wants students to grasp how recently humans appeared. The class textbook lists the eras on one page with equal spacing. What design best addresses the goal?

- A. Copying the textbook's equal-spaced list onto the tape.
- B. Memorizing the names and order of the eras.
- C. Any long timeline will always show scale.
- D. A timeline drawn to scale, where each meter stands for the same number of years, with human history as a tiny sliver at the end.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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2. [TRANSFER — reversed stem / different variable]

On a 46-meter scaled timeline of Earth's 4.6-billion-year history, how far from the present end is the start of the dinosaurs, about 230 million years ago?

- A. 2.3 meters
- B. 23 meters
- C. 46 meters
- D. 43.7 meters

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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3.

Students used a stream table with sand and a garden hose to model river erosion. The table was donated by a local engineering firm. The water ran for ten minutes and carved a meandering channel. What is the strongest follow-up task?

- A. Tell students real rivers form meanders in exactly ten minutes.
- B. Conclude the model is a perfect copy of a real river.
- C. Have students list ways the stream table differs from a real river, such as time, sediment type and scale.
- D. Write a thank-you letter to the engineering firm.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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4.

A student who loves astronomy and has a telescope at home writes: “It’s summer in July because Earth is closer to the Sun then.” The class is in the Northern Hemisphere. Which fact most directly shows the student’s idea is wrong?

- A. Telescopes show the Sun is very far away.
- B. Earth’s orbit is a perfect circle.
- C. The Sun is hotter in July.
- D. Earth is closest to the Sun in early January, when it is winter in the Northern Hemisphere.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

When it is summer in the Northern Hemisphere, what season is it in the Southern Hemisphere, and why?

- A. Winter, because the Southern Hemisphere is tilted away from the Sun.
- B. Summer, because the whole Earth is closer to the Sun.
- C. Winter, because the Southern Hemisphere is farther from the Sun.
- D. Summer, because both hemispheres always share the same season.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Using laptops from a recently upgraded cart, students download a month of earthquake locations from a public geological survey feed, plot them on a world map, and notice that most fall along plate boundaries. Which practice are they mainly using?

- A. Patterns, which is the practice they used.
- B. Using technology, which is the ninth practice.
- C. Analyzing and interpreting data.
- D. Obtaining, evaluating and communicating information.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a lab report written on the school's new learning platform, a student claims the rock sample is sedimentary. Evidence: it contains rounded grains cemented together and a small shell fossil. Reasoning: "The rock has rounded grains and a fossil." What is wrong with the argument?

- A. The claim is missing.
- B. The report should have been handwritten.
- C. The reasoning only restates the evidence; it should explain that sedimentary rocks form from cemented sediments where organisms can be buried and preserved.
- D. The evidence is too weak; a fossil never appears in sedimentary rock.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A student's reasoning reads: "Because warm air rises and cools, water vapor condenses, so clouds form over the heated land." Which part of claim-evidence-reasoning is this?

- A. A hypothesis, which is a separate fourth part.
- B. Reasoning.
- C. Claim.
- D. Evidence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Before a unit on the water cycle, a teacher in her third year at the school gives a short probe asking where the water on the outside of a cold glass comes from. Many students say it leaks through the glass. The school requires two grades per week. How should she use the probe results?

- A. Skip the water cycle, since students already have an explanation.
- B. Use the answers to plan activities that test the leaking idea against evidence, such as a glass of colored water.
- C. Record the results as the unit's summative assessment.
- D. Grade the probe to meet the two-grades rule.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A teacher gives the same ungraded condensation probe at the end of the unit that she gave at the start. What can the two sets of answers show that the first set alone could not?

- A.** Proof that every student has mastered the water cycle.
- B.** How students' ideas changed during instruction, and which misconceptions persisted.
- C.** Students' ideas before instruction.
- D.** A grade for the unit, since both probes can be averaged.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher plans a field trip to a road-cut outcrop to study rock layers. Parents have volunteered to drive, and the site has great views of the valley. What must be done before the trip?

- A.** Visit or assess the site for hazards such as traffic and falling rock, get approvals and permission forms, and set supervision and an emergency plan.
- B.** Collect permission slips; nothing else is required.
- C.** Tell students on arrival to stay away from anything that looks dangerous.
- D.** Plan the lookout stop so students can photograph the valley.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a mineral identification lab with a new set of specimen boxes, students test hardness with a steel nail, check streak on a porcelain plate, and place one drop of dilute hydrochloric acid on each sample to test for calcite. What eye protection is required?

- A.** Gloves only, since one drop cannot reach the eyes.
- B.** Goggles only for the streak test.
- C.** None, because most steps are harmless.
- D.** Chemical splash goggles for the whole lab, because of the acid step.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An earth science teacher ~~with a long hallway and a roll of adding machine tape~~ wants students to grasp how recently humans appeared. The class textbook lists the eras on one page with equal spacing. What design best addresses the goal?

Target word: design that shows deep time **Decides:** Intervals must be proportional to show scale.

A	Dropped element	Leaves out proportion, which is the whole point.
B	Adjacent category	Sequence is not scale.
C	Absolute language	Length alone does not; the intervals must be proportional.
D	Correct	Proportional intervals make the scale visible.

2. Answer A Concept 1 • transfer

On a 46-meter scaled timeline of Earth's 4.6-billion-year history, how far from the present end is the start of the dinosaurs, about 230 million years ago?

Target word: scaled distance **Decides:** $46 \text{ m} / 4.6 \text{ billion years} = 1 \text{ m per } 100 \text{ million years}$.

A	Correct	$230 \text{ million} / 100 \text{ million per meter} = 2.3 \text{ m}$.
B	Inverted operation	Off by a factor of ten; uses 10 million years per meter.
C	Wrong stem number	The full length of the timeline.
D	Inverted operation	Measures from the start of Earth instead of from the present.

3. Answer C Concept 2

Students used a stream table with sand and a garden hose to model river erosion. ~~The table was donated by a local engineering firm.~~ The water ran for ten minutes and carved a meandering channel. What is the strongest follow-up task?

Target word: follow-up for model use **Decides:** Students should evaluate the model's limits.

A	Wrong stem number	Takes the model's run time as real-world time.
B	Absolute language	All models simplify.
C	Correct	Evaluating a model's limits is part of using models well.
D	Camouflage	Courteous, but not a science follow-up.

4. Answer D Concept 3

A student ~~who loves astronomy and has a telescope at home~~ writes: "It's summer in July because Earth is closer to the Sun then." The class is in the Northern Hemisphere. Which fact most directly shows the student's idea is wrong?

Target word: evidence against the distance idea **Decides:** Earth is actually closest to the Sun in early January, during Northern Hemisphere winter.

A	Camouflage	Answers the telescope detail; distance alone does not explain seasons.
B	Absolute language	False; the orbit is slightly elliptical.
C	Inverted operation	Accepts a cause that does not exist; the Sun's output does not change by season.
D	Correct	If distance caused seasons, January would be summer in the north.

5. Answer A Concept 3 • transfer

When it is summer in the Northern Hemisphere, what season is it in the Southern Hemisphere, and why?

Target word: season and cause in the other hemisphere **Decides:** Tilt points one hemisphere toward the Sun while the other points away.

A	Correct	Opposite tilt gives opposite seasons.
B	Confusion set	Uses the distance misconception.
C	Adjacent category	Right season, wrong cause.
D	Absolute language	Seasons are opposite across the equator.

6. Answer C Concept 4

Using laptops ~~from a recently upgraded cart~~, students download a month of earthquake locations from a public geological survey feed, plot them on a world map, and notice that most fall along plate boundaries. Which practice are they mainly using?

Target word: which practice **Decides:** Plotting real data to find a pattern.

A	Adjacent category	Patterns is a crosscutting concept, not a practice.
B	Camouflage	Technology is a tool; it is not a practice itself.
C	Correct	They organize and look for patterns in real measurements.
D	Confusion set	That practice centers on reading and judging sources, not finding patterns in data.

7. Answer C Concept 5

In a lab report ~~written on the school's new learning platform~~, a student claims the rock sample is sedimentary. Evidence: it contains rounded grains cemented together and a small shell fossil. Reasoning: "The rock has rounded grains and a fossil." What is wrong with the argument?

Target word: weakness in the argument **Decides:** The reasoning repeats the evidence instead of linking it to a principle.

A	Dropped element	The claim is present: the rock is sedimentary.
B	Camouflage	The platform does not decide the quality of the argument.
C	Correct	Reasoning connects evidence to the claim with a scientific principle.
D	Absolute language	Fossils are most often found in sedimentary rock.

8. Answer B Concept 5 • transfer

A student's reasoning reads: "Because warm air rises and cools, water vapor condenses, so clouds form over the heated land." Which part of claim-evidence-reasoning is this?

Target word: which part **Decides:** It uses a scientific principle to link evidence to claim.

A	Adjacent category	Hypotheses belong to investigation design, not the CER frame.
B	Correct	It explains why, using the principle of rising, cooling air.
C	Confusion set	The claim is the answer to the question, stated without the why.
D	Confusion set	Evidence is the observation or data, not the principle.

9. Answer B Concept 6

Before a unit on the water cycle, a teacher in her third year at the school gives a short probe asking where the water on the outside of a cold glass comes from. Many students say it leaks through the glass. ~~The school requires two grades per week.~~ How should she use the probe results?

Target word: use of a pre-instruction probe **Decides:** Probes reveal ideas to plan around; they are not scored for correctness.

A	Inverted operation	The probe shows a misconception, which calls for more instruction.
B	Correct	Formative probes shape instruction around students' actual ideas.
C	Adjacent category	Summative assessment comes at the end, not before teaching.
D	Camouflage	Grading pre-instruction ideas penalizes students for not yet being taught.

10. Answer B Concept 6 · transfer

A teacher gives the same ungraded condensation probe at the end of the unit that she gave at the start. What can the two sets of answers show that the first set alone could not?

Target word: value of a pre/post comparison **Decides:** Comparing before and after reveals change in students' ideas.

A	Absolute language	A short probe gives evidence, not proof of full mastery.
B	Correct	The first probe is the baseline; the second shows change against it.
C	Dropped element	The first probe alone already shows that.
D	Adjacent category	Probes are formative; they are not meant as summative grades.

11. Answer A Concept 7

A teacher plans a field trip to a road-cut outcrop to study rock layers. Parents have volunteered to drive, and ~~the site has great views of the valley.~~ What must be done before the trip?

Target word: pre-trip requirement **Decides:** A prior hazard assessment of the site, plus approvals and an emergency plan.

A	Correct	Hazards are identified and controlled before students arrive.
B	Absolute language	Permission forms are one part, not the whole plan.
C	Dropped element	Leaves out the advance hazard assessment and shifts judgment to students.
D	Camouflage	Answers the view detail, not safety.

12. Answer D Concept 8

In a mineral identification lab ~~with a new set of specimen boxes,~~ students test hardness with a steel nail, check streak on a porcelain plate, and place one drop of dilute hydrochloric acid on each sample to test for calcite. What eye protection is required?

Target word: required eye protection **Decides:** The acid drop is the most hazardous step.

A	Absolute language	Splashes and rubbed eyes can still expose the eyes.
B	Confusion set	Picks the wrong step as the hazard.
C	Adjacent category	Matches PPE to the typical step, not the most hazardous.
D	Correct	PPE is set by the riskiest step; acid can splash.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Scale, proportion and quantity (deep time)	Vast times and distances become meaningful only through scaled models where intervals are proportional, such as a geologic timeline drawn to scale.	Using an equal-spaced list of eras, which hides that most of Earth's history came before complex life.	Millions or billions of years, or distances too large to picture.
2	Models have limits	Every model simplifies; a strong lesson has students use a model and then name what it gets wrong or leaves out.	Treating a classroom model as an exact copy of the Earth system.	A globe, lamp, stream table or diagram used to explain a process.
3	Cause of the seasons	Seasons are caused by the tilt of Earth's axis, which changes the angle and length of daily sunlight; distance from the Sun is not the cause.	Believing summer happens when Earth is closest to the Sun.	A student explanation that mentions near or far.
4	Analyzing and interpreting data with real datasets	Using real-time or archived datasets (earthquakes, weather, stream gauges) to find patterns is the practice of analyzing and interpreting data, and a strong reason to use technology.	Calling any use of a website the practice of obtaining information, or using the data only to look things up.	Students graph, map or look for patterns in actual measurements.
5	Claim, evidence, reasoning	In argument from evidence, the reasoning explains why the evidence supports the claim, using a scientific principle.	Letting the reasoning simply restate the evidence or the claim.	A student argument where one part is missing or repeats another.
6	Eliciting preconceptions before instruction	Short formative probes given before instruction reveal students' existing ideas so the teacher can plan around them; they are not graded for correctness.	Grading the pre-assessment or skipping it because the topic is familiar.	The start of a unit and a question about what to do first.
7	Field-trip safety planning	Outdoor and field work requires a prior site visit or hazard assessment, approvals and permission forms, adequate supervision, and an emergency plan before students go.	Relying on students to judge hazards at the site, or planning safety on the day.	An outcrop, stream or quarry visit being planned.
8	The most hazardous step sets the PPE	Choose protective equipment for the most hazardous step of a lab, even if most steps are harmless; for example, a drop of dilute acid on minerals requires splash goggles.	Matching PPE to the typical step instead of the riskiest one.	A mostly harmless lab with one chemical or heat step.

Course scope (public description): Science Methods—Secondary Earth Science focuses on teaching methods specific to science for graduate students seeking an endorsement in secondary earth science. Course content focuses on the design and teaching of standards-based lessons using the three dimensions of science (science and engineering practices, crosscutting concepts, and disciplinary core ideas) and the appropriate integration of

technology into those lessons. Students in this course work within their content areas to evaluate, enhance, and plan appropriate science instruction. This course includes laboratory safety training and certification, which includes safe laboratory practices and procedures for science classrooms and the proper use of personal protective equipment. Previous coursework in curriculum, instruction, and assessment is a prerequisite for this course.